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1. Introduction to PC Hardware

A personal computer (PC) is made up of several hardware components that work together to process, store, and display information. Understanding each component, its function, and how they interconnect is the foundation of computer hardware maintenance.

This assignment covers the complete practical assessment for PC Hardware including identification of components, step-by-step assembly and disassembly, wired network configuration, hardware troubleshooting, and preventive maintenance.

What is Computer Hardware?

Computer hardware refers to the physical parts of a computer system. Unlike software (which consists of programs and data), hardware is tangible — you can see and touch it. Hardware components are classified into:

Category	Examples	Function
Input Devices	Keyboard, Mouse, Scanner	Send data to the computer
Output Devices	Monitor, Printer, Speakers	Display processed results
Processing Unit	CPU, GPU	Execute instructions
Storage Devices	HDD, SSD, USB Drive	Store data permanently
Memory	RAM, ROM, Cache	Temporary working storage
Connectivity	NIC, Wi-Fi Card, Modem	Connect to networks

Importance of PC Hardware Knowledge

- Helps diagnose and fix computer problems faster
- Enables upgrading components to improve performance
- Reduces dependency on costly technical support
- Essential for IT support and network administration careers
- Required knowledge for BCA, BSc CSIT, and BIT programs at TU

2. Safety & Anti-Static Precautions

Before touching any internal PC component, it is critical to follow proper safety procedures. Electronic components are extremely sensitive to static electricity, physical shock, and incorrect handling.

Electrostatic Discharge (ESD)

Electrostatic Discharge (ESD) is the sudden flow of electricity between two objects caused by contact. A discharge of even 10 volts can damage a microchip, while humans cannot feel anything below 3,000 volts. This means you can damage a component without even knowing it.

Anti-Static Safety Rules

DO	DO NOT
Wear anti-static wrist strap	Touch components without grounding
Work on hard, flat surface	Work on carpeted floor
Ground yourself first (touch PC case)	Wear wool or synthetic clothing
Store parts in anti-static bags	Place parts on plastic surfaces
Handle PCBs by the edges	Touch IC chips directly
Turn off PC and unplug before work	Work on powered PC
Use proper screwdriver size	Force connectors that don't fit

Required Safety Equipment

- Anti-static wrist strap (ESD strap)
- Anti-static mat for workbench
- Non-magnetic Phillips head screwdrivers (size 1 and 2)
- Cable ties and labeling tape
- Compressed air can for cleaning
- Isopropyl alcohol (90% or higher) for cleaning

Section 9.1 – Hardware Identification & Handling

The first practical task requires students to correctly identify, label, and explain the function of each major PC component. You must also demonstrate proper anti-static handling techniques.

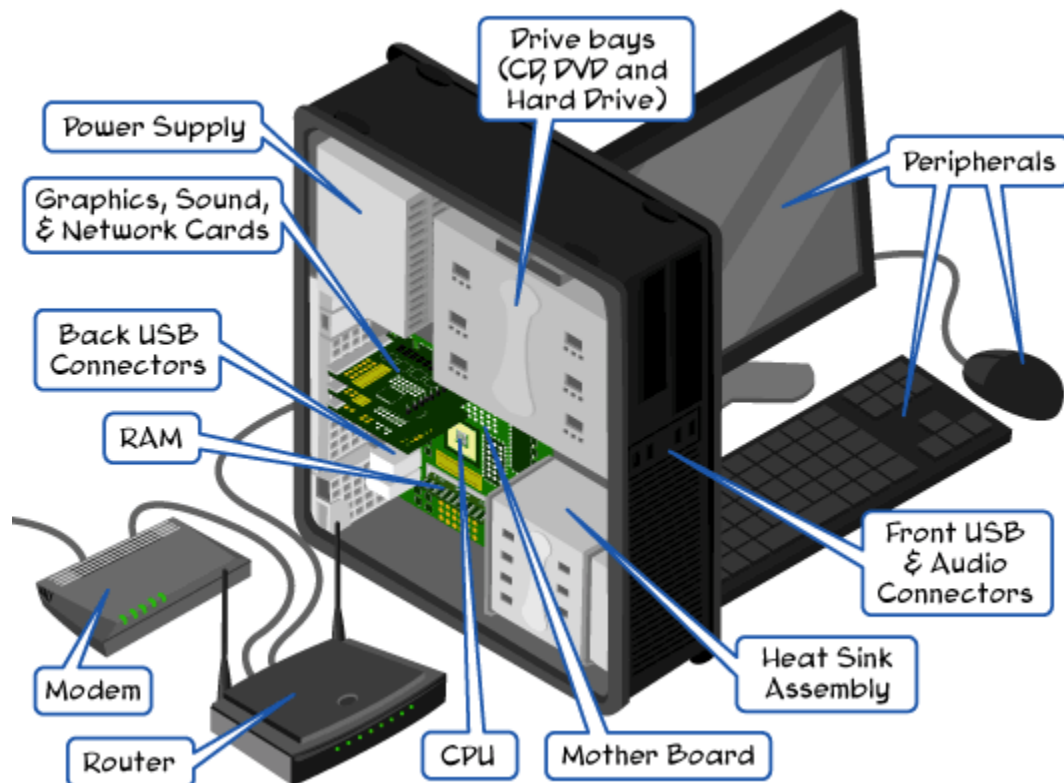


Figure 1: Main PC Components and Their Locations on a Motherboard

Component Identification Table

Component	Full Name	Function	Location
CPU	Central Processing Unit	Executes all instructions and calculations	CPU socket on motherboard
RAM	Random Access Memory	Temporary memory for running programs	RAM slots on motherboard
ROM	Read Only Memory	Stores BIOS/UEFI firmware permanently	Chip on motherboard

HDD	Hard Disk Drive	Magnetic permanent storage	Drive bay, SATA cable
SSD	Solid State Drive	Fast flash-based permanent storage	Drive bay or M.2 slot
PSU	Power Supply Unit	Converts AC power to DC voltages	Top or bottom of case
GPU	Graphics Processing Unit	Renders images, video, and 3D graphics	PCIe x16 slot
NIC	Network Interface Card	Connects computer to LAN/internet	PCIe slot or onboard
Motherboard	Main Circuit Board	Connects and communicates with all parts	Screwed inside the case
SMPS	Switch Mode Power Supply	Efficient power conversion unit	Same as PSU
BIOS Chip	Basic Input/Output System	Initializes hardware at startup	Small chip near CMOS battery

Motherboard Form Factors

- ATX (Advanced Technology Extended) – Most common; 305 x 244 mm
- Micro-ATX (mATX) – Smaller; 244 x 244 mm, fewer expansion slots
- Mini-ITX – Very compact; 170 x 170 mm, for small form factor builds
- E-ATX (Extended ATX) – Larger server/workstation boards

CPU Socket Types

- Intel LGA 1700 – For 12th/13th Gen Intel Core processors
- Intel LGA 1200 – For 10th/11th Gen Intel processors
- AMD AM5 – For Ryzen 7000 series processors
- AMD AM4 – For Ryzen 1000-5000 series processors

4. Section 9.2 – PC Assembly & Disassembly

PC Assembly is the process of installing all hardware components into a computer case to build a working PC. Disassembly is the reverse process. Both must be done carefully following the correct order.

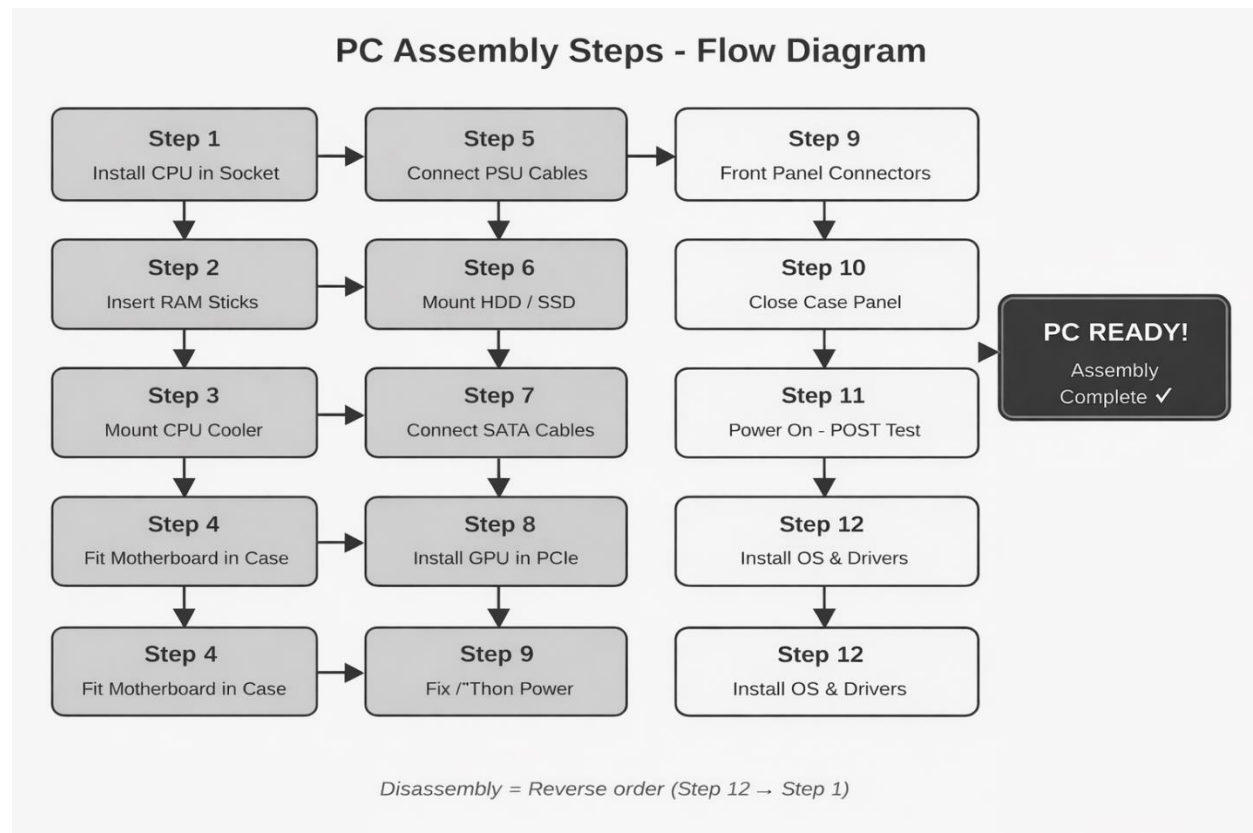


Figure 2: PC Assembly Steps – Complete Flow Diagram (Follow steps 1–12 in order)

Detailed Assembly Steps

Step 1: Prepare the Workspace

- Clear a large, flat, non-carpeted workspace
- Gather all tools: screwdrivers, anti-static strap, cable ties
- Wear anti-static wrist strap connected to a grounded surface
- Open the PC case by removing the side panel screws

Step 2: Install the CPU

1. Open the CPU socket lever on the motherboard

2. Align the CPU with the notches/triangle marker on the socket
3. Carefully lower the CPU – DO NOT apply pressure
4. Close the socket lever to lock the CPU in place

Step 3: Install RAM

5. Check motherboard manual for correct dual-channel slots (usually A2+B2)
6. Open the clips on both sides of the RAM slot
7. Align the RAM notch with the slot key
8. Press down firmly until both clips click into place

Step 4: Apply Thermal Paste & Install CPU Cooler

9. Apply a pea-sized amount of thermal paste to the center of the CPU
10. Place the cooler on top and align the mounting holes
11. Screw in the cooler diagonally (corner to corner) for even pressure
12. Connect the fan cable to the CPU_FAN header on the motherboard

Step 5: Mount Motherboard in Case

13. Install I/O shield into the back of the case first
14. Align motherboard with standoff screws in the case
15. Screw in the motherboard – do not overtighten

Steps 6–12: Connecting Everything

Step	Action	Details
6	Connect PSU cables	24-pin ATX power + 8-pin EPS CPU power
7	Install HDD/SSD	Mount in drive bay, connect SATA data + power
8	Install GPU	Slot into PCIe x16, connect 6/8-pin power cable
9	Front panel connectors	Power SW, Reset SW, HDD LED, Power LED, USB header
10	Connect case fans	To SYS_FAN headers or fan hub
11	Close the case	Replace side panel and secure screws
12	First power on	Listen for 1 beep = success. No beep = check connections

Disassembly Order

Disassembly is done in REVERSE order of assembly. Follow these key rules:

- Always shut down and unplug the PC before opening
- Remove GPU and expansion cards first
- Disconnect all cables before removing the motherboard
- Label cables and take photos before disconnecting for reassembly reference
- Store screws in labelled containers or magnetic tray

5. Section 9.3 – Network Setup (Wired LAN)

A Local Area Network (LAN) connects multiple computers within a limited area such as a lab, office, or home. In this practical, you will configure a wired Ethernet connection between PCs using static IP addressing.

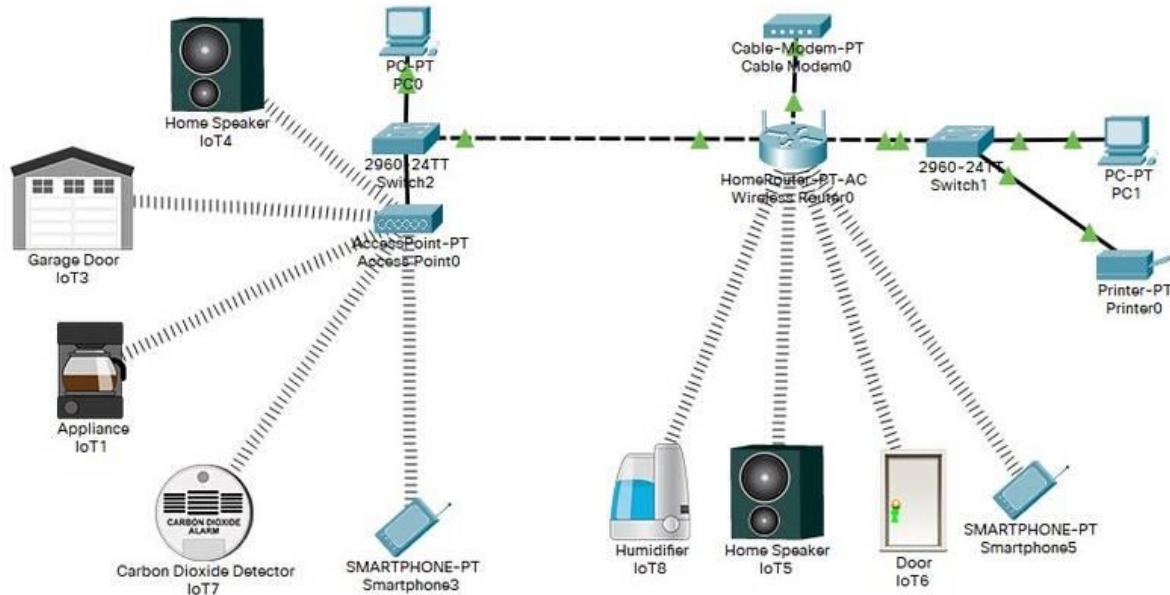


Figure 3: Wired LAN Network Diagram – Computers connected through a Switch/Router

Network Concepts & Terminology

Term	Explanation
IP Address	A unique numerical label assigned to each device on the network (e.g., 192.168.1.10)
Subnet Mask	Defines the network range. 255.255.255.0 means the first 3 numbers identify the network
Gateway	The IP address of the router that connects local network to the internet (e.g., 192.168.1.1)
DNS	Domain Name System – converts website names to IP addresses
MAC Address	Physical hardware address of the network card – unique per device
DHCP	Automatic IP assignment by the router (Dynamic Host Configuration Protocol)

ping	Command to test network connectivity by sending packets to another IP address
LAN Cable	Cat5e or Cat6 Ethernet cable used to physically connect devices

5.2 Step-by-Step: Wired LAN Configuration

Step 1: Physical Connection

16. Get a Cat5e or Cat6 RJ45 Ethernet cable
17. Plug one end into the PC's Ethernet/NIC port
18. Plug the other end into an available port on the switch or router
19. Verify the link light (usually green/amber LED) lights up on both ends

Step 2: Configure Static IP (Windows)

20. Press Windows + R, type 'ncpa.cpl' and press Enter
21. Right-click on 'Ethernet' or 'Local Area Connection'
22. Select 'Properties'
23. Double-click on 'Internet Protocol Version 4 (TCP/IPv4)'
24. Select 'Use the following IP address'
25. Enter the IP settings shown in the table below
26. Click OK on all windows to save

Setting	PC 1 (Your Computer)	PC 2 (Lab Computer)
IP Address	192.168.1.10	192.168.1.11
Subnet Mask	255.255.255.0	255.255.255.0
Default Gateway	192.168.1.1	192.168.1.1
Preferred DNS	8.8.8.8 (Google DNS)	8.8.8.8 (Google DNS)

Step 3: Test Connection with Ping

27. Press Windows + R, type 'cmd' and press Enter
28. Type: ping 192.168.1.1 (tests router connection)
29. Type: ping 192.168.1.11 (tests connection to another PC)
30. If you see 'Reply from...' messages – connection is successful!
31. If you see 'Request timed out' – check cable and IP settings

6. Section 9.4 – Troubleshooting Simulations

Hardware troubleshooting is the process of diagnosing and fixing problems with PC components. In the lab, you will use POST diagnostic cards, interpret beep codes, and use command-line tools to identify and resolve issues.

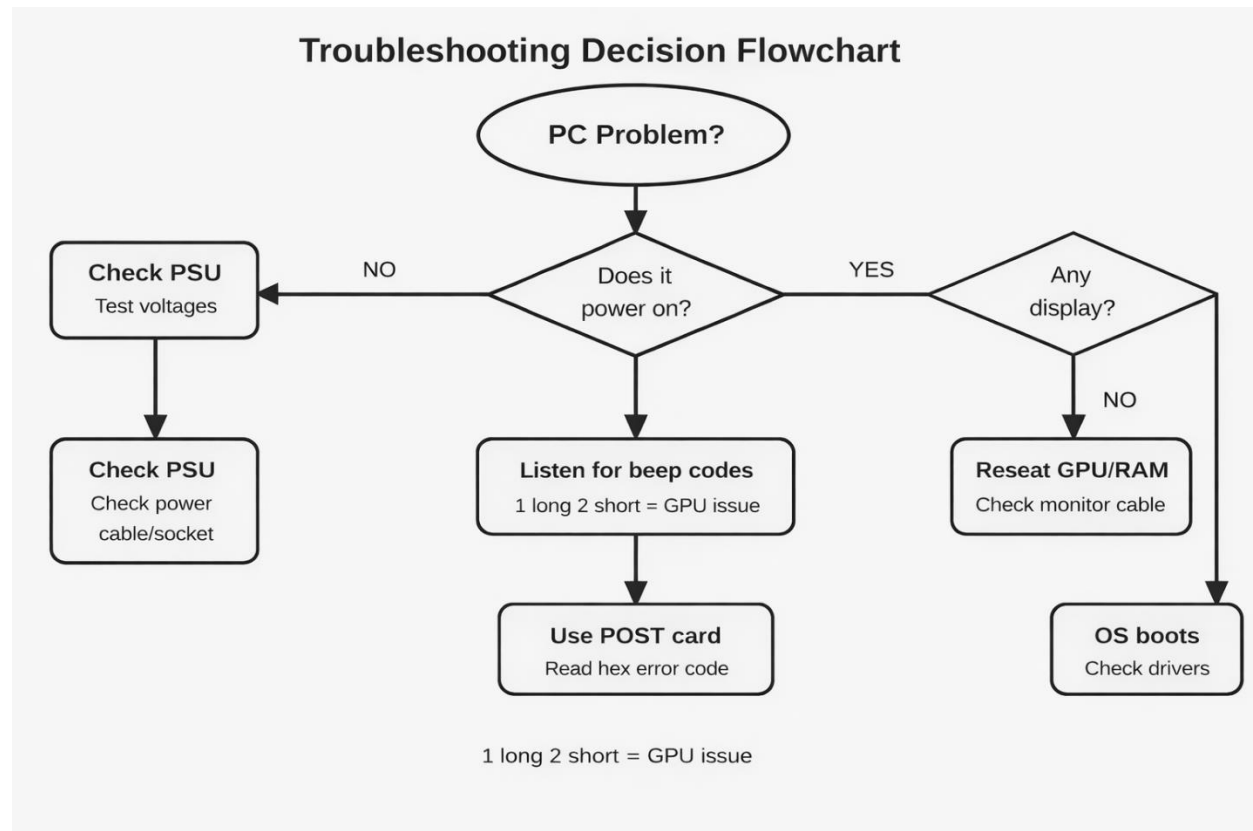


Figure 4: Hardware Troubleshooting Decision Flowchart

POST (Power-On Self-Test)

When you press the power button, the BIOS performs a POST to check all hardware. If a problem is found, it signals through beep codes or a POST card code.

Common BIOS Beep Codes (AMI BIOS)

Beep Pattern	Component	Solution
1 Short Beep	Normal POST – All OK	No action needed
2 Short Beeps	POST Error / CMOS issue	Check CMOS battery, reset BIOS
1 Long Beep	Memory (RAM) error	Reseat RAM, try one stick at a time
1 Long + 2 Short	GPU / Display error	Reseat GPU, check monitor cable
1 Long + 3 Short	Video memory error	Replace GPU or check VRAM
Continuous Beeps	RAM not seated properly	Remove and firmly reseat all RAM
No Beep, No Boot	PSU or motherboard failure	Test PSU with tester, check connections

Useful Diagnostic Commands (Windows CMD)

Command	Purpose	Example
ping	Test network connectivity	ping 192.168.1.1
ipconfig	Show IP configuration	ipconfig /all
tracert	Trace route to a destination	tracert google.com
netstat	Show active connections	netstat -an
chkdsk	Check disk for errors	chkdsk C: /f
sfc /scannow	Scan system files for corruption	sfc /scannow
msinfo32	View complete system information	msinfo32
devmgmt.msc	Open Device Manager	devmgmt.msc

7. Section 9.5 – Maintenance Tasks

Regular preventive maintenance extends the life of a PC and prevents unexpected failures. In this section, you will learn to clean internal components, apply thermal paste correctly, and verify the power supply.

Cleaning Internal Components

Dust accumulation is one of the most common causes of PC overheating and failure. Regular cleaning every 3–6 months is recommended.

Component	Cleaning Method	Frequency
CPU Heatsink & Fan	Compressed air + soft brush	Every 3 months
Case Fans	Compressed air, wipe blades	Every 3 months
Motherboard	Soft brush + compressed air	Every 6 months
RAM Slots	Compressed air only	Every 6 months
GPU Fan & Heatsink	Compressed air + brush	Every 3 months
PSU vents	Compressed air from outside	Every 6 months
Case interior	Vacuum + microfiber cloth	Every 3 months

Thermal Paste Application

Thermal Paste Application Guide

STEP 1
Remove old paste



CPU Surface

Use isopropyl alcohol 90%+
Lint-free cloth

STEP 2
Apply new paste



Pea-sized dot
in the CENTER
Do NOT spread!

STEP 3
Attach cooler



Press cooler down
evenly on CPU
Screw in diagonally

RESULT
Even spread



Paste spreads
automatically
under pressure ✓

Warning: Too much paste causes overheating! Too little = poor contact.

Ideal CPU temperature: 30-50°C idle | 60-80°C under load

Figure 5: Thermal Paste Application – Step-by-Step Guide

Thermal paste (also called thermal compound) fills microscopic air gaps between the CPU and the heatsink, improving heat transfer dramatically.

When to Replace Thermal Paste:

- Every 2–3 years as part of routine maintenance
- When CPU temperatures are unusually high (above 85°C under load)
- After removing the CPU cooler for any reason
- When the old paste has dried out and turned to powder

Step-by-Step Thermal Paste Replacement:

32. Power off the PC and unplug the power cable completely
33. Remove the CPU cooler by unscrewing in a diagonal pattern
34. Apply a few drops of isopropyl alcohol (90%+) on a lint-free cloth
35. Gently wipe off old thermal paste from CPU and cooler base
36. Let the surfaces dry completely (1–2 minutes)
37. Apply a pea-sized dot of new thermal paste to the center of CPU
38. Reattach the cooler firmly – the paste will spread under pressure
39. Screw in diagonally for even pressure distribution
40. Connect CPU fan cable to CPU_FAN header
41. Power on and monitor CPU temperature using HWMonitor or CPU-Z

Power Supply Verification

The PSU provides clean, regulated power to all components. A failing PSU can cause random shutdowns, data corruption, and component damage.

Voltage Rail	Rated Voltage	Acceptable Range	Powers
+12V	12.0V	11.4V – 12.6V	CPU, GPU, fans, drives
+5V	5.0V	4.75V – 5.25V	USB ports, older drives
+3.3V	3.3V	3.135V – 3.465V	Motherboard logic circuits
-12V	-12.0V	-10.8V – -13.2V	Serial ports (legacy)
+5VSB	5.0V (standby)	4.75V – 5.25V	Wake-on-LAN, USB charging

Other Preventive Maintenance Tasks

- Check all cable connections for looseness every 6 months
- Run MemTest86+ for RAM diagnostics every few months
- Monitor CPU/GPU temperatures under normal and load conditions
- Check HDD health using CrystalDiskInfo or HDDScan
- Update drivers, BIOS, and operating system regularly
- Defragment HDD (not SSD) quarterly
- Check for loose screws that may cause short circuits

CONCLUSION

This lab report provided comprehensive practical knowledge of PC hardware and system maintenance. Throughout the lab activities, we learned how to identify different computer components and understand their functions in a complete system.

We followed proper safety and anti-static precautions, which are essential to protect sensitive electronic components from damage. The process of PC assembly and disassembly helped us understand how each hardware part is connected and installed correctly.

Additionally, we successfully configured a wired LAN network using static IP addressing and tested connectivity using command-line tools like ping. The troubleshooting section enhanced our ability to diagnose hardware and system problems using POST beep codes and system commands.

Finally, we performed various maintenance tasks such as cleaning internal components, applying thermal paste, and checking power supply conditions. These practices are important to ensure system stability, performance, and longevity.

Overall, this lab has improved our practical skills and confidence in handling computer hardware, which is essential for future studies and careers in IT and computer science.